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asteroids = [5, 10, -5]

Cases:-

collision tabhi ho sakta hai jab:-

Condition | $\text{stack.top} > 0 \ \&\& \ \text{current asteroid} < 0$ |

ex:-

case I 5 -3 $\Rightarrow$ collision
 $\rightarrow$ $\leftarrow$

case II -5 -3 $\Rightarrow$ No collision,
 $\leftarrow$ $\leftarrow$

case III. -5 3 $\Rightarrow$ No collision
 $\leftarrow$ $\rightarrow$

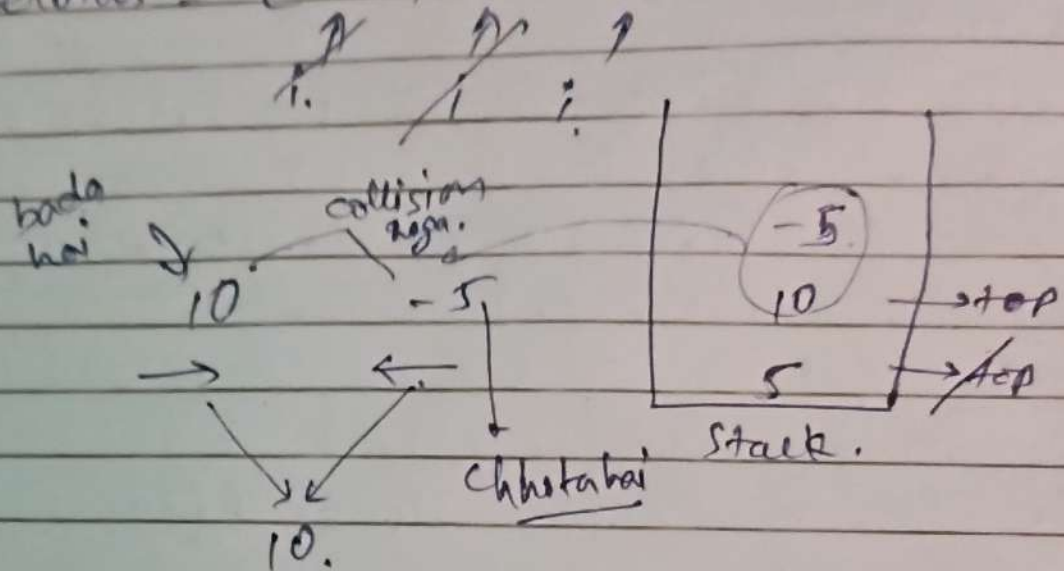
Ans

while (!stack.isEmpty()) &&
stack.peek() > 0 &&
asteroid < 0

↳ This code of this.

dry run:-

asteroids = [5, 10, -5]

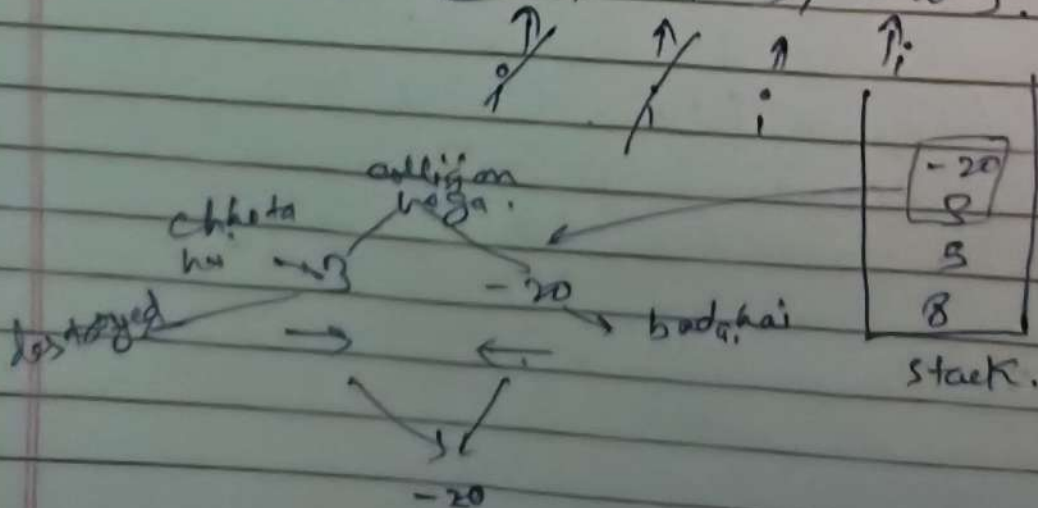


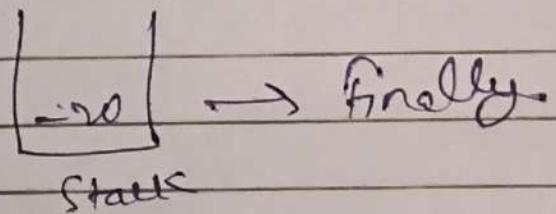
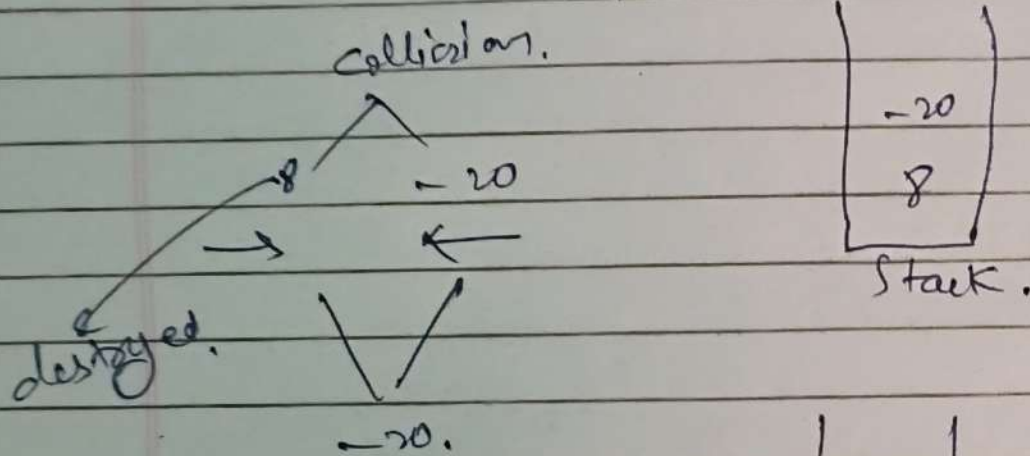
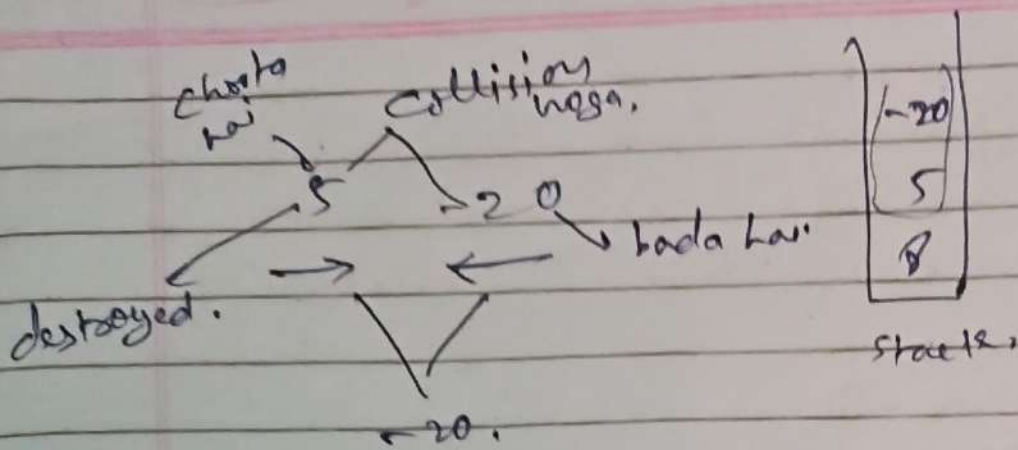
i.e -5 vanish ho jayega.

or stack ek ander [10, 5] bachega.

another examples:-

asteroids = [8, 5, 3, -20]





Note:-

→ While is used because one asteroid can collide with multiple asteroids already present in stack.

After destroying one asteroid, the current asteroid may still be alive, so we must check again.

This creates a chain reactions.